3.5 Line and Rotational Symmetry

A figure has line symmetry if some reflection maps it onto itself, and rotational symmetry if some rotation less than 360° does the same — and these two properties are surprisingly independent, since a shape can have one without the other. This section explains *why* a regular n-gon always has exactly n lines of symmetry and rotational order n, using the Section 3.4 theorem that two reflections compose into a rotation, and it highlights the classic trap that a non-square rectangle's diagonals are not lines of symmetry at all.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Lines of symmetry of equilateral triangle?

2. Lines of symmetry of a square?

3. Lines of symmetry of non-square rectangle?

4. Lines of symmetry of isosceles (not equilateral)?

5. Lines of symmetry of scalene triangle?

6. Least positive rotational angle for square?

7. Least positive rotational angle for equilateral triangle?

8. Order of rotational symmetry?

9. Lines of symmetry?

10. Lines of symmetry of a circle?

11. Lines of symmetry of a non-rhombus parallelogram?

12. Lines of symmetry of a non-square rhombus?

13. A figure mapped to itself by 180° but no smaller positive angle has order?

14. A regular octagon has how many lines of symmetry, and what is its smallest positive angle of rotational symmetry?

15. A shape has exactly 2 lines of symmetry crossing at 90°. Using the Section 3.4 composition theorem, what rotational symmetry angle must the shape also have?

✓ Answer Key

1. 3
2. 4
3. 2
4. 1
5. 0
6. 90°
7. 120°
8. 5
9. 6
10. Infinitely many
11. 0
12. 2
13. 2
14. 8 lines; 45°
15. 180° (composing the two reflections gives a rotation of $2 \times 90° = 180°$)